

basic-digital Optimized tmux — User & Operator Guide

Status: Phase 38, Bug #33 (this session, 2026-05-07). **Constitution alignment:** §11.4 anti-bluff covenant (gated PATH export), §12.9 containerized build pattern, §12.10 continuation-doc invariant.

§1 What this is

A reproducible, containerized build of `tmux 3.6a` (latest stable) with:

1. **Hardened compile flags** — `-O2 -DNDEBUG -fstack-protector-strong -D_FORTIFY_SOURCE=2, RELRO` + immediate-binding link.
2. **Build-time `-ljemalloc`** — jemalloc directly linked at the binary level (more aggressive RAM return than glibc malloc).
3. **Runtime `LD_PRELOAD=libjemalloc.so`** — additional safety; ensures jemalloc engages even on hosts where the linker resolved a different malloc.
4. **OOM-score protection** — wrapper script sets `oom_score_adj=-500` on the spawned server, making tmux survive nearly all OOM cascades (preserving operator session for investigation).
5. **Bounded history-limit** — explicit `2000` (the default; making it explicit prevents accidental future bumps).
6. **Hermetic install** — built artifact lives in `<project>/tmux/build/`. `PATH` export points there; system tmux untouched.

Why this is gated by §11.4 — the original article warns about "apparent leaks" in tmux. Our forensic record (§12 incidents 2026-04-27, 2026-04-28, 2026-04-30, 2026-05-06) shows tmux was *not* the actual cause of any documented host-distress event. Building from source therefore would not have prevented those incidents. We do this work for **defensive** reasons (so future unforeseen tmux issues never affect us) and we GATE the install behind a full test suite to make sure the new build is genuinely better than the system tmux — not silently worse.

§2 One-command install

```
# Step 0 – install build deps (one-time, requires sudo)
sudo bash scripts/install_deps.sh

# Step 1 – build + verify + install (no sudo)
bash scripts/setup.sh
```

The orchestrator runs five gated phases:

Phase	Action	Gate
1	Verify host build deps present	aborts if gcc / pkg-config libevent missing
2	Containerized build (podman, mem_limit=2g)	aborts if container cannot be built or fails to produce tmux/build/bin/tmux
3	Generate tmx wrapper script (LD_PRELOAD + OOM score)	aborts if wrapper template missing
4	Verification gate — run all 14 tests via verify.sh	STOPS HERE if any FAIL — no PATH export
5	Install ~/.tmux.conf + ~/.bashrc snippet	only runs if Phase 4 GREEN

§3 Architecture

```

<project>/
├─ tmux/                                     # Git submodule
(upstream tmux/tmux, pinned to 3.6a)
├   └─ build/bin/tmux                       # Built artifact
(volume-mounted from container)
├─ docker/
├   └─ Dockerfile.tmux-build                # Ubuntu 22.04 +
libevent-dev + libncurses-dev + libjemalloc-dev
├   └─ docker-compose.tmux-build.yml        # mem_limit:2g, cpus:2,
network:none
├   └─ build_inside_container.sh             # Runs autogen +
configure + make + install (in-container)
└─ scripts/
    └─ install_deps.sh                      # Host package
installer (sudo, OS-aware)
    └─ build_containerized.sh               # Orchestrator that
runs the container
    └─ tmux.conf.template                  # Optimized config
template
    └─ tmx.template                        # Wrapper template
(LD_PRELOAD + OOM + scope + colour)
    └─ bashrc_snippet.template             # PATH-export snippet
for ~/.bashrc
    └─ hostname_color.sh                   # Deterministic
hostname → status-bg colour
    └─ tests/
        └─ 01_smoke.sh                     # Binary
version
    └─ 02_session.sh                       # new-session/
list-sessions/kill-server
    └─ 03_jemalloc_loaded.sh               # /proc/PID/
maps grep for jemalloc
    └─ 04_history_limit.sh                 # show-options
after config

```

```

    |   |— 05_clear_history_releases.sh           # The article's
"apparent leak" check
    |   |— 06_concurrent_panes.sh               # 10-pane RSS
bounding
    |   |— 07_long_session.sh                   # 30-s
sustained activity
    |   |— 08_oom_score_adj.sh                   # /proc/PID/
oom_score_adj == -500
    |   |— 09_crash_isolation_scope.sh           # cgroup-v2
transient scope readbacks + crash containment
    |   |— 10_hostname_color_algorithm.sh        # DJB2 hash →
palette determinism + spread
    |   |— 11_hostname_color_integration.sh      # wrapper
applies expected status-bg
    |   |— 12_memory_pressure_under_cap.sh       # destructive:
alloc up to MemoryMax, OOM-kill evidence
    |   |— 13_tasksmax_stress.sh                 # destructive:
fork-bomb resistance via TasksMax
    |   |— 14_concurrent_oom_independence.sh     # destructive:
kill scope A, B + C survive
    |   |— meta_test_false_positive_proof.sh     # §11.4.4
layer-4 paired-mutation harness
    |   |— run_all.sh                           # Orchestrator
    |— challenges/
    |   |— tmux.yaml                            # HelixQA Challenges
(mirrors tests with §11.4 evidence semantics)
    |— verify.sh                               # Gate that decides
green/red
    |— setup.sh                                # Master orchestrator
(one command)

```

§4 The 14 verification tests — what they prove

Test	Proves	If FAIL
01 smoke	binary invocable, version matches pin	broken build
02 session	basic lifecycle works	broken libevent integration
03 jemalloc loaded	LD_PRELOAD engages at runtime	wrapper broken or jemalloc absent — fall back to glibc malloc (not catastrophic)
04 history-limit	user config respected	broken option- parsing
05 clear-history releases	the article's "apparent leak" — clear-history actually frees memory	glibc fragmentation (no jemalloc) — degrades to WARN, not FAIL

Test	Proves	If FAIL
06 concurrent panes	10 panes don't OOM the server	severe leak
07 long session	25 s sustained activity grows < 50%	gradual leak
08 oom_score_adj	wrapper applies -500	wrapper script broken — biggest §12-protection benefit gone
09 crash isolation scope	cgroup-v2 transient scope enforces MemoryMax/CPUQuota/TasksMax; SIGKILL containment does not take user.slice down	per-session isolation broken
10 hostname colour algorithm	DJB2 → 27-palette deterministic on same hostname; spread ≥ 12/16 across distinct names	host distinguishability broken
11 hostname colour integration	wrapper applies expected status-bg to running server; persists on second session	colour-on-attach broken
12 memory pressure under cap (destructive, TMX_TEST_DESTRUCTIVE=1)	allocation up to MemoryMax triggers OOM-kill of scope only; user.slice survives	memory cap not enforced
13 TasksMax stress (destructive)	fork-bomb caps at TasksMax=4096; cgroup pids interface readback	task-count cap not enforced
14 concurrent OOM independence (destructive)	kill scope A → scopes B and C survive with original MainPIDs	scope independence broken

Plus a §11.4.4 layer-4 paired-mutation harness (`meta_test_false_positive_proof.sh`) with 6 registered mutations against tests 09 / 10. The gate is considered self-validating only when all mutations are caught.

Gate logic (`scripts/tests/run_all.sh + scripts/verify.sh`): every test is treated equally. Any test line starting with FAIL aborts the gate (exit 1, no PATH export). Tests that emit a SKIP line are counted as SKIP and do not block (intended for honest precondition gates — see Constitution §11.4.2). Tests 12 / 13 / 14 are destructive and opt-in via `TMX_TEST_DESTRUCTIVE=1` — they SKIP by default on non-dedicated hosts. There is no per-test "blocker / critical / advisory" hierarchy in the gate code: the rule is "any FAIL = RED".

§5 Operator commands

Goal	Command
Install everything (Linux)	<code>sudo bash scripts/install_deps.sh && bash scripts/setup.sh</code>
Install everything (macOS)	<code>brew install podman && podman machine init && podman machine start && bash scripts/setup.sh</code>
Re-verify after upstream pull (Linux)	<code>bash scripts/verify.sh</code>
Re-verify after upstream pull (macOS)	<code>bash scripts/test_vm.sh</code>
Run full destructive suite	<code>TMX_TEST_DESTRUCTIVE=1 bash scripts/test_vm.sh</code>
Run §11.4.4 paired-mutation meta-test	<code>META=1 bash scripts/test_vm.sh</code>
Force rebuild	<code>bash scripts/setup.sh --rebuild</code>
Uninstall	<code>bash scripts/setup.sh --uninstall</code>
Run a specific test	<code>TMUX_BIN=\$(pwd)/tmux/build/bin/tmux bash scripts/tests/03_jemalloc_loaded.sh</code>
Update tmux to a new pinned tag	<code>cd tmux && git checkout <new-tag> && cd .. && bash scripts/setup.sh --rebuild</code>

§5.6 Native dual-OS per-session isolation (architecture since 2026-05-13)

Each `tmx new -s NAME` invocation produces its own tmux server running as a **host-native process** with OS-specific resource isolation. The session's shell IS the operator's host shell — same `$USER`, same `$HOME`, same `$PATH`, full filesystem access. The wrapper applies the strongest containment each OS natively supports.

Mechanism per OS

	Linux	macOS (Darwin)
Binary format	ELF 64 (built via <code>build_containerized.sh</code> or <code>build_native.sh</code>)	Mach-O 64 (built via <code>build_native.sh</code> against Homebrew deps)
Output dir	<code>tmux/build/bin/tmux</code>	<code>tmux/build-darwin/bin/tmux</code>
Isolation primitive	cgroup-v2 transient scope via <code>systemd-run --user --scope --unit=tmx-NAME.scope</code>	POSIX rlimit wrapper (<code>scripts/tmx-rlimit-wrapper.sh</code>) applied as session default-command

	Linux	macOS (Darwin)
Memory cap	MemoryMax (kernel-enforced per cgroup)	NOT ENFORCED — see "Honest gap" below
CPU cap	CPUQuota=200% (per cgroup)	RLIMIT_CPU (kernel-enforced per process, SIGXCPU on hard exhaust)
Task cap	TasksMax=4096 (per cgroup)	RLIMIT_NPROC (kernel-enforced per user)
jemalloc preload	LD_PRELOAD	DYLD_INSERT_LIBRARIES + DYLD_FORCE_FLAT_NAMESPACE=1
OOM containment	OOM in scope ⇒ kernel kills only that scope; user.slice survives	CPU/NPROC enforced; memory: process may grow until host swaps
OOM helper	tmx-oom-set (setcap, sets oom_score_adj=-500)	N/A (no oom_score_adj interface on Darwin)

Honest gap (macOS memory cap)

Per Constitution §1 / §11.4 anti-bluff: **the Darwin XNU kernel does NOT enforce RLIMIT_AS / RLIMIT_DATA / RLIMIT_RSS for unprivileged processes.** Verified:

```
$ ulimit -v 102400 # try to cap virtual memory at 100 MB
bash: ulimit: virtual memory: cannot modify limit: Invalid argument

$ python3 -c "x = 'a' * 200_000_000; print(len(x))"
alloc OK: 200000000 # process allocates 200 MB freely
```

The tmx-rlimit-wrapper on Darwin therefore applies ONLY the limits the kernel actually enforces (RLIMIT_CPU, RLIMIT_NPROC). The setup.sh closing message documents this explicitly. For full memory containment on macOS, you'd need launchd jobs with HardResourceLimits (root) or — for the same workload on a Linux host — cgroup scopes which DO enforce memory.

Naming + sanitisation (both OSes)

- **Socket:** tmx-<sanitised-NAME> (under /tmp/tmux-<uid>/). Tmux routing via -L tmx-NAME.
- **Scope unit (Linux only):** tmx-<sanitised-NAME>.scope. Operator-targetable: systemctl --user status tmx-mywork.scope.
- **Sanitisation:** characters outside [A-Za-z0-9._-] → _. Collision (scope already active OR server already running on socket) errors out explicitly.

Caps (per session, both OSes)

Cap	Default	Override	Linux	Darwin
Memory	host-adaptive: max(MemTotal)	TMX_MEM=8G tmx new -s heavy		

Cap	Default	Override	Linux	Darwin
	× 60% / 4, 2 GB)		✓ cgroup MemoryMax enforced	× NOT enforced (XNU gap)
CPU	200% (2 cores) on Linux; 86400 s (24 h) on Darwin	TMX_CPU=400 (Linux); TMX_CPU_HARD_SEC=3600 (Darwin)	✓ CPUQuota	✓ RLIMIT_CPU
Tasks	4096	(env: no; edit template)	✓ TasksMax per scope	✓ RLIMIT_NPROC per user

Cleanup

tmx kill-session -t NAME:

1. tmx -L tmx-NAME kill-session -t NAME
2. **Linux only:** systemctl --user stop tmx-NAME.scope (explicit scope teardown for cgroup reclaim)

tmx kill-server enumerates every /tmp/tmux-<uid>/tmx-* socket, kills each server, and (Linux) stops every tmx-*.scope unit.

Verifying isolation (positive evidence, both OSes)

Linux:

```
$ tmx new -s a -d; tmx new -s b -d
$ systemctl --user list-units --type=scope --no-legend | grep tmx-
tmx-a.scope loaded active running ...
tmx-b.scope loaded active running ...
$ cat /sys/fs/cgroup/.../tmx-a.scope/memory.max
2147483648 ← per-session memory
cap
$ cat /sys/fs/cgroup/.../tmx-a.scope/cgroup.procs
<session A PIDs> ← distinct from B's
```

Darwin:

```
$ tmx new -s a -d; tmx new -s b -d
$ tmx ls
a: 1 windows (created ...)
b: 1 windows (created ...)
$ # Inside session a (via tmx attach):
$ ulimit -t ; ulimit -u
86400 ← RLIMIT_CPU enforced
2666 ← RLIMIT_NPROC
enforced
$ id ; hostname ; which brew
uid=501(milosvasic) gid=20(staff) ... ← operator's host user
Mistborn.local ← operator's macOS
```

host
/opt/homebrew/bin/brew ← Homebrew reachable

Test 15 + e2e T7 in scripts/tests/ automate these checks with OS-dispatched assertions per Constitution §11.4.2.

§5.7 Clipboard — native selection by default, tmux mouse on demand, paste-IN

The shipped `~/.tmux.conf` (generated from `scripts/tmux.conf.template`) defaults to **set -g mouse off**, so the **terminal owns the mouse**: native click-drag selection (including multi-line), right-click → Copy, and native scroll all work — identically on Linux and macOS, on every emulator (iTerm2, Terminal.app, GNOME Terminal, WezTerm, ...), inside or outside a full-screen TUI such as Claude Code / HelixCode. This is the recommended copy/paste path. **prefix m** enables tmux's own mouse on demand (wheel scrollbar inside TUIs + drag-copy to the OS clipboard); **prefix P** pastes the OS clipboard into a pane.

Why the default is `mouse off`: a wire-level test (test 59) proved that with `mouse on` tmux emitted mouse-tracking DECSET enables (`CSI ?1000h` / `?1002h` / `?1006h`) that **suppressed** the emulator's native selection and right-click → Copy — the root cause of the long-standing "can't select / copy" reports. With `mouse off` tmux emits none of those enables, so the native mouse is unobstructed.

Quick reference

Goal	Keystroke / mouse
Select & copy anything (default) — plain shell, Claude Code, vim, less, htop	native click-drag (multi-line OK) → <code>Cmd-C</code> / right-click → Copy
Paste OS clipboard INTO the pane	native <code>Cmd-V</code> / right-click → Paste, or <code>prefix + P</code> (capital — lowercase is previous-window)
Wheel-scroll tmux scrollbar inside a TUI, or tmux drag-copy to clipboard	prefix + m (toggle tmux mouse ON) → wheel / drag; <code>prefix + m</code> again to return to native
Keyboard-only copy	<code>prefix + [→ v → arrows</code> / <code>j / k → y</code>

The mechanisms

1. **mouse off default + prefix m toggle** — `set -g mouse off` ships as the default so the terminal's native selection / right-click → Copy / scroll are unobstructed (no mouse-tracking DECSET enables emitted). `bind m set -g mouse \;` `display-message '...'` toggles tmux mouse ON on demand for wheel scrollbar inside a TUI and tmux drag-copy, and back OFF to native. (tmux toggles a flag/choice option when its value is omitted.)

2. **@clip user option** — OS-adaptive WRITE pipeline (pbcopy → wl-copy → xclip → termux-clipboard-set → OSC-52 fallback). copy-pipe-and-cancel "#{@clip}" routes every tmux y / Enter / MouseDragEnd1Pane selection through it (used in the mouse on state).
3. **@clip-read user option** — symmetric READ pipeline (pbpaste → wl-paste → xclip -o → termux-clipboard-get → empty). Drives prefix + P.
4. **Modifier-drag overrides (fallback for the mouse on state only)** — bind -n M-MouseDrag1Pane copy-mode -M + bind -n S-MouseDrag1Pane copy-mode -M. When tmux mouse is ON and a TUI also captures mouse events (Claude Code's alt-screen + DECSET 1002/1003), the unmodified MouseDrag1Pane forwards the drag to the app via send -M. Holding Alt / Option (macOS) or Shift (Linux) prefixes the event with M- / S- and tmux routes it to copy-mode regardless of the app's mouse mode. Not needed in the default mouse off state, where native selection works with no modifier.
5. **Bracketed-paste on paste-IN** — bind P run -b 'sh -c "pbpaste 2>/dev/null || wl-paste -n 2>/dev/null || xclip -o -selection clipboard 2>/dev/null || termux-clipboard-get 2>/dev/null" | tmux load-buffer - && tmux paste-buffer -p'. The -p flag (per man tmux: *"paste bracket control codes are inserted around the buffer if the application has requested bracketed paste mode"*) makes shells / editors treat the inserted block as literal text — no accidental command execution from a stray newline.

Verification (positive evidence per §101)

Test	What it proves	Mutation
44 — clipboard copy-OUT physical	y keystroke → @clip → pbpaste round-trip with unique marker	M44 strips @clip definition
45 — multi-line keyboard copy	v + arrows + y selects ≥3 lines; round-trip via pbpaste	—
46 — paste-IN physical	prefix + P reads @clip-read and inserts into the pane	M46 strips @clip-read
47 — modifier mouse surface	tmux observes M- / S- modifier on drag-start	—
48 — modifier-drag binding chain	M-MouseDrag1Pane + M-MouseDragEnd1Pane resolves through @clip end-to-end	M48 strips the modifier binding

The synthetic alt-screen surrogate at `scripts/tests/helpers/synthetic_alt_screen_app.py` substitutes for the Claude Code CLI in tests 47 / 48, avoiding §11.4.98 OAuth/interactive flake while still exercising the exact alt-screen + mouse-tracking surface.

The dedicated operator-recipe document is [docs/guides/clipboard.md](#) — quick recipes (native selection by default + prefix m on demand), prefix P paste-IN, troubleshooting (empty @clip-read, scrolling tmux history inside a TUI, the mouse on modifier-drag fallback).

Sources verified 2026-06-13

- **tmux upstream man page** — <https://man.openbsd.org/tmux.1> (re-verified 2026-06-13 via WebFetch; OpenBSD ships the canonical upstream `tmux.1`). Confirms `set-clipboard external` "will attempt to set the terminal clipboard but ignore attempts by applications to set tmux buffers" (the OSC-52 copy-OUT path) and that a flag/choice option set with its value omitted toggles its value (the prefix `m / set -g mouse toggle`). The `mouse off` $\Rightarrow$ no mouse-tracking DECSET enables behaviour (so native selection / right-click $\rightarrow$ Copy / scroll is unobstructed) is proven at the wire level in this repo by test 59 (`scripts/tests/59_*.sh`), the load-bearing authority for the default-architecture claims above. Canonical binding source: `scripts/tmux.conf.template`.

Sources verified 2026-05-28

- **tmux 3.6a man page** — `man tmux` on the host. Confirms `S- / M-` modifier prefixes apply to mouse events (`MouseDownPane` is a key name; key names accept `C- / S- / M-` prefixes per the "KEY BINDINGS" section); confirms `paste-buffer -p` enables bracketed-paste; confirms `@`-prefixed user options accept arbitrary string values.
 - **tmux upstream man page mirror** — <https://man.openbsd.org/tmux.1> (OpenBSD ships the canonical upstream `tmux.1`).
 - **Anthropic Claude Code docs** — <https://code.claude.com/docs/en/> (verified 2026-05-28; no tmux-specific integration page is published; the alt-screen + mouse-tracking behaviour cited above is documented from direct observation against Claude Code CLI v2.x).
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§6 Why we did this despite the data

The forensic record (§12 incidents) does NOT name tmux as a root cause. The actual culprits were `soong_build`, `kotlinc`, `gradle`, `git pack-objects` — all already addressed by §12.6/§12.7/§12.8/§12.9. So the question naturally arises: why this work?

Three reasons:

1. **Defense-in-depth.** §12 incidents are rare but catastrophic when they happen. `tmux` survives at `oom_score_adj=-500` even if the `user.slice` goes down — meaning the operator can attach to the surviving session and investigate, instead of losing all context.
2. **Reproducibility.** The system `tmux` on a future host might be 1.x with known leaks. Pinning to 3.6a + `jemalloc` removes that variable.
3. **Documentation as a teaching artifact.** The 14 tests + Challenges show *exactly* what "tmux is healthy" means in measurable terms. Future engineers can re-run these on any new host.

This is justified Defense-in-Depth, not a fix for an existing problem. The investigation explicitly recorded that the article's premise doesn't apply to us; we still proceeded because the marginal cost is small and the upside is non-zero.

§8 Full OOM protection options (CAP_SYS_RESOURCE required)

oom_score_adj=-500 requires CAP_SYS_RESOURCE, which a regular user shell does NOT have. The Linux kernel rejects writes of negative oom_score_adj values from non-root processes (per Documentation/filesystems/proc.rst).

The wrapper (tmx) attempts the write defensively but the kernel will silently no-op for non-root operators. Test 08 SKIPS honestly to acknowledge this rather than reporting a misleading PASS.

Three real options for full OOM protection

Option A — Run wrapper via sudo (simplest, annoying UX)

```
alias tmux='sudo -E /path/to/tmx'
```

Pro: works immediately, no new binaries. Con: requires sudo password every tmux launch.

Option B — setcap-enabled oom_set helper (recommended)

Build a tiny C helper that has CAP_SYS_RESOURCE via setcap, called by the wrapper:

```
/* oom_set.c - minimal helper to set oom_score_adj on a target PID
   */
#include <stdio.h>
#include <stdlib.h>
int main(int argc, char **argv) {
    if (argc != 3) return 1;
    char path[256];
    snprintf(path, sizeof path, "/proc/%s/oom_score_adj",
             argv[1]);
    FILE *f = fopen(path, "w");
    if (!f) return 2;
    fputs(argv[2], f);
    fclose(f);
    return 0;
}
```

Build + grant capability (one-time):

```
gcc -o oom_set oom_set.c
sudo setcap cap_sys_resource+ep ./oom_set
sudo install -m 755 ./oom_set /usr/local/bin/tmx-oom-set
```

Then update the wrapper to call tmx-oom-set "\$server_pid" -500 instead of writing directly.

Pro: no per-invocation sudo. Helper has only one capability (minimal blast radius).
Con: requires one-time root setup; the helper binary must be audited (it's tiny so this is easy).

Option C — systemd user service with `OOMScoreAdjust=`

Run tmux as a systemd user service whose unit file declares `OOMScoreAdjust=-500`. systemd applies the adjustment at process startup (it has `CAP_SYS_RESOURCE`).

```
# ~/.config/systemd/user/tmux.service
[Service]
ExecStart=/path/to/tmux new-session -A -s main
OOMScoreAdjust=-500
Restart=on-failure

[Install]
WantedBy=default.target
```

```
systemctl --user enable --now tmux.service
```

Pro: clean, capability-less from the operator's perspective. systemd handles everything. Con: tmux runs in a different cgroup hierarchy; some plugins / `$TMUX_PANE` integrations may behave differently.

Recommendation

For most operators: **Option C** if you're already on a systemd-based distro and use tmux as your "main" workspace. **Option B** if you launch tmux ad-hoc from terminals. **Option A** if you only want OOM protection during high-risk tasks (a single sudo-tmux session for the AOSP build window).

For most production use, none of these is mandatory. The `oom_score_adj` work is the smallest of the three benefits we get from the new build (the bigger ones being jemalloc + bounded history-limit + reproducible 3.6a pin). The covenant tests SKIP cleanly and the binary is GREEN as-is.

§7 Cross-references

- CLAUDE.md — Applied Fixes Reference, Phase 38
- docs/CONTINUATION.md §3 — current Phase 38 status
- docs/guides/ATMOSPHERE_CONSTITUTION.md §11.4, §12.9 — invoked patterns
- tools/helixqa/HelixQA/banks/atmosphere.yaml — TMUX-CH-01..08 entries